

AI AGENT SYSTEMS

Module #1 – From Chatbots to Multiagent Systems

Outline

- Agenthood, Agency and Agentic AI
- Agent-Based Modeling
- Methods for Agent-Based Modeling
- Generative AI and Large Language Models (LLMs)
- Case Study: Agentic Trading Systems
  - Task: Should I Trade QQQ Today?
  - Workflow: Automating Routine Tasks
  - Agent: Autonomous Decision-Making
  - Multi-Agent Systems: Specialization and Coordination



Agenthood - General Definitions

Merriam-Webster Dictionary:

> "... one that acts or exerts power... a means or instrument by which a guiding intelligence achieves a result"

Russell and Norving

> "An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors."



Agenthood – More Specific Definitions

Maes, Pattie:

> "Autonomous Agents are computational systems that inhabit some complex dynamic environment, sense and act autonomously in this environment, and by doing so realize a set of goals or tasks for which they are designed".

Smith, Cypher and Spohrer:

> " a persistent software entity dedicated to a specific purpose. 'Persistent' distinguishes agents from subroutines; agents have their own ideas about how to accomplish tasks, their own agendas. 'Special purpose' distinguishes them from multifunction applications; agents are typically much smaller.

Agenthood – Industrial Definition

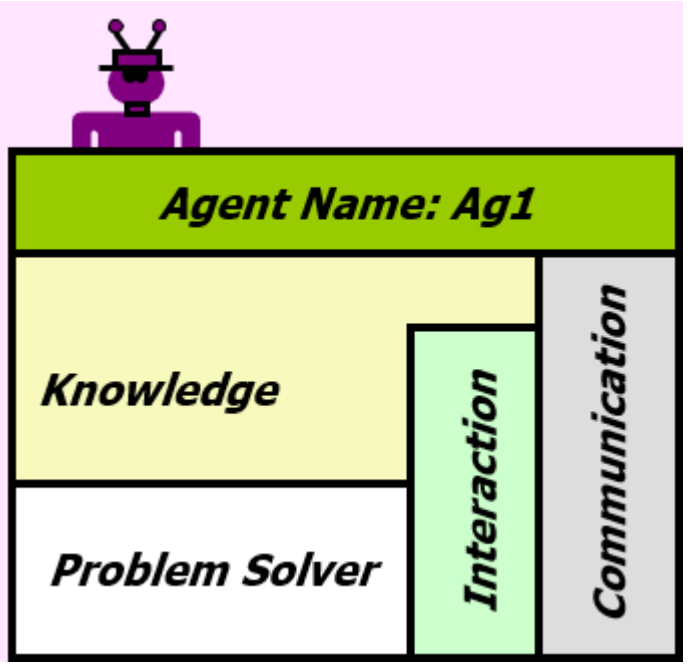
IBM:

> "Intelligent agents are software entities that carry out some set of operations on behalf of a user or another program with some degree of independence or autonomy, and in doing so, employ some knowledge or representations of the user's goals or desires"

Agenthood-
CIR- Agent Architecture

An agent is an individual collection of primitive components

- The basic components include:
  - knowledge
  - problem-solving
  - interaction
  - communication



Agency
in the context of Agentic AI

Agency: The capacity of an agent or AI system to autonomously select and execute actions in pursuit of goals, based on its observations, reasoning, and feedback from the environment.

Term	Main question	Meaning in agentic AI
Agency	“How much can this system act on its own?”	The capacity to make decisions, select actions, pursue goals, use tools, respond to feedback, and influence an environment

Agentic AI – More Specific Definitions

Stanford University Human-Centered Artificial Intelligence

Agentic AI refers to AI systems designed to act as autonomous or semi-autonomous agents: they can set or interpret goals, plan and sequence actions, use tools (like web browsers, code, or APIs), make decisions based on feedback, and adapt over time to complete tasks. Unlike a purely reactive chatbot that only responds turn-by-turn, **agentic AI** is oriented around ongoing task execution—breaking down objectives, coordinating steps, and sometimes operating with minimal human oversight within defined constraints

What is Agentic AI? | Stanford HAI

Agentic AI – More Specific Definitions

A recent ACM survey characterizes an LLM-based agent as an LLM combined with components that allow it to **perceive, interact with, and modify an external environment**.

OpenAI's engineering definition is useful as a complementary reference. It defines agents as systems that can independently accomplish tasks on a user's behalf, with an LLM controlling workflow execution and selecting tools dynamically.

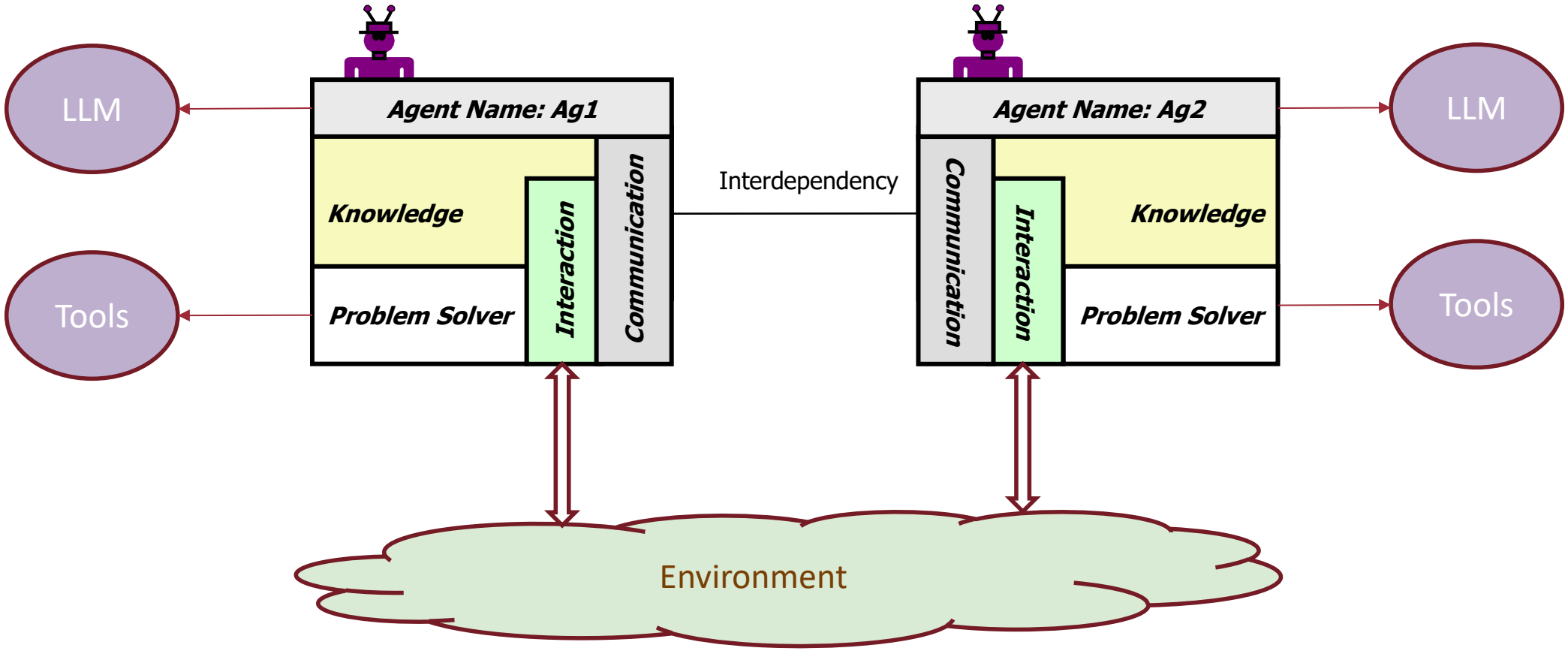
A practical guide to building agents | OpenAI

Agentic AI –
LLM ≠ AI agent ≠ AI Agent Systems

A simple conceptual progression is:

- **LLM**
  → generates/reasons about information
- **AI agent**
  → LLM + goal + instructions + tools + action control loop
- **AI Agent Systems**
  → agents that autonomously pursue goals through repeated **observe → reason/plan → act → observe feedback → adapt**

Agent- Agency – Agentic AI – Multiagent Systems



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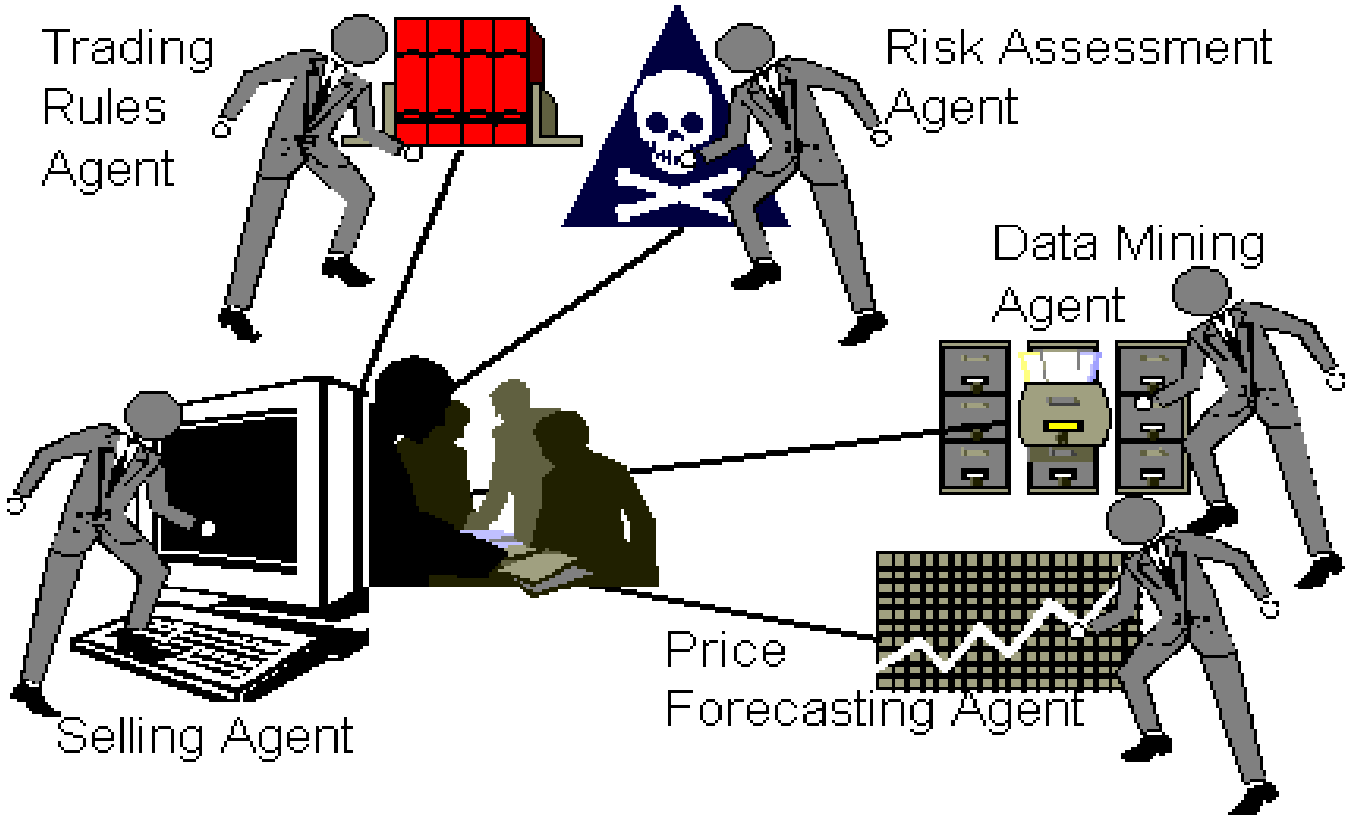
What is the Appropriate Modeling Paradigm for Systems ?

- Functional-Oriented
  - The modeling process is information driven. Information flow characteristic is used to derive system structure



- Object-Oriented
  - Modeling a system by assembling communities of interoperable objects
- Agent-Oriented
  - Modeling a system by assembling communities of interacting agents

Multiagent Trading System



ABM: Structure and Purpose

- Modelling systems composed of autonomous, interacting agents
- Include models of behavior
  - Human or otherwise, can be described by simple rules, or sophisticated AI models.
  - Interactions with other agents influence their behaviors.
- Used to observe the collective effects of agent behaviors and interactions.

ABM: Distinguishing Features

- By modelling agents individually, the full effects of the diversity that exists among agents in their attributes and behaviors can be observed as it gives rise to the behavior of the system as a whole.
- Modelling systems from the 'ground up', agent- by-agent and interaction-by-interaction—self-organization can often be observed in such models. Patterns, structures, and behaviours emerge that were not explicitly programmed into the models but arise through the agent interactions.

ABM for Social Simulation

- Agent-based modelling offers a way to model social systems that are composed of agents who interact with and influence each other, learn from their experiences, and adapt their behaviours so they are better suited to their environment
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ABM for Social Simulation

A social agent-based model can be used to analyze a social system consisting of

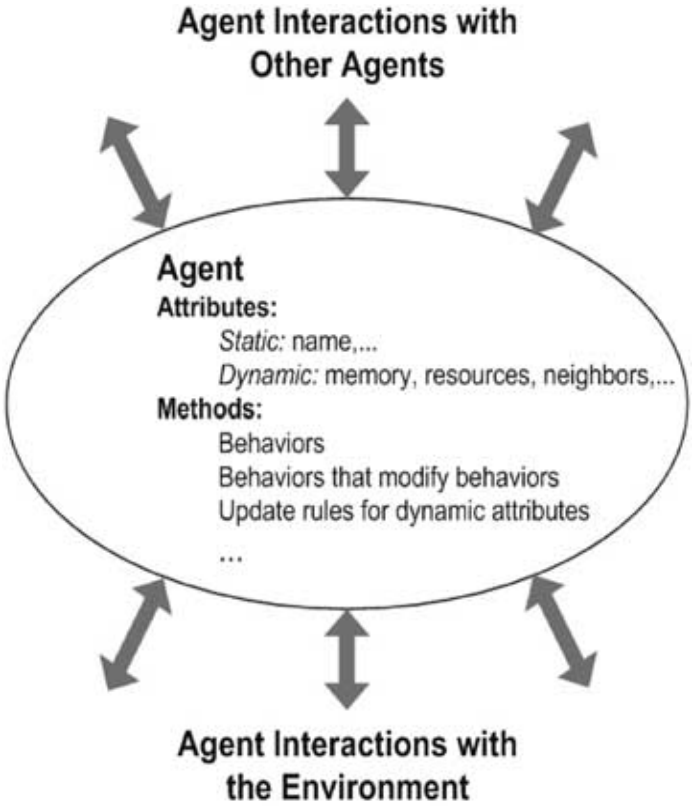
- autonomous, interacting, goal-oriented, bounded-rational set of *actors* that use
- a given *interaction rule set* and are situated in
- an *environment*"

Structure of an Agent-Based Model

The main result of the modeling is a conceptual and formal model of the referent social system specified by

- Agents (attributes and behaviors),
- Interaction protocols (methods) and
- Agents' environment
  - where agents are situated and
  - interact with their environment in addition to other agents.

A model developer must identify, model, and program these elements to create an agent-based model



Agents in ABM

- **Human actors** in an ABM—whether individuals or collectives (e.g., households, groups, other social aggregates)—are represented as agents that encapsulate attributes and behaviors (computational methods or operations).
- The **state of an agent** is determined by its attributes

Agents in ABMS: Behavior

- Agents are endowed with behaviors that allow them to make independent decisions.
- Behaviors can be derived from simplistic and reactive 'if-then' rules to complex behaviors modelled by adaptive artificial intelligence techniques, such as neural networks.
- Behaviors can be adaptive: An agent can learn and change its behaviors in response to its experiences using adaptive mechanisms



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Methods for agent-based modelling

1. What specific problem should be solved by the model? What specific questions should the model answer? What value-added would agent-based modelling bring to the problem that other modelling approaches cannot bring?
2. What should the agents be in the model? Who are the decision makers in the system? What are the entities that have behaviours? What data on agents are simply descriptive (static attributes)? What agent attributes would be calculated endogenously by the model and updated in the agents (dynamic attributes)?

Methods for agent-based modelling Cont.

3. What is the agents' environment? How do the agents interact with the environment? Is an agent's mobility through space an important consideration?
4. What agent behaviours are of interest? What decisions do the agents make? What behaviours are being acted upon? What actions are being taken by the agents?

Methods for agent-based modelling Cont.

5. How do the agents interact with each other? With the environment? How expansive or focused are agent interactions?
6. Where might the data come from, especially on agent behaviours, for such a model?
7. How might you validate the model, especially the agent behaviours?

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Generative AI Definition

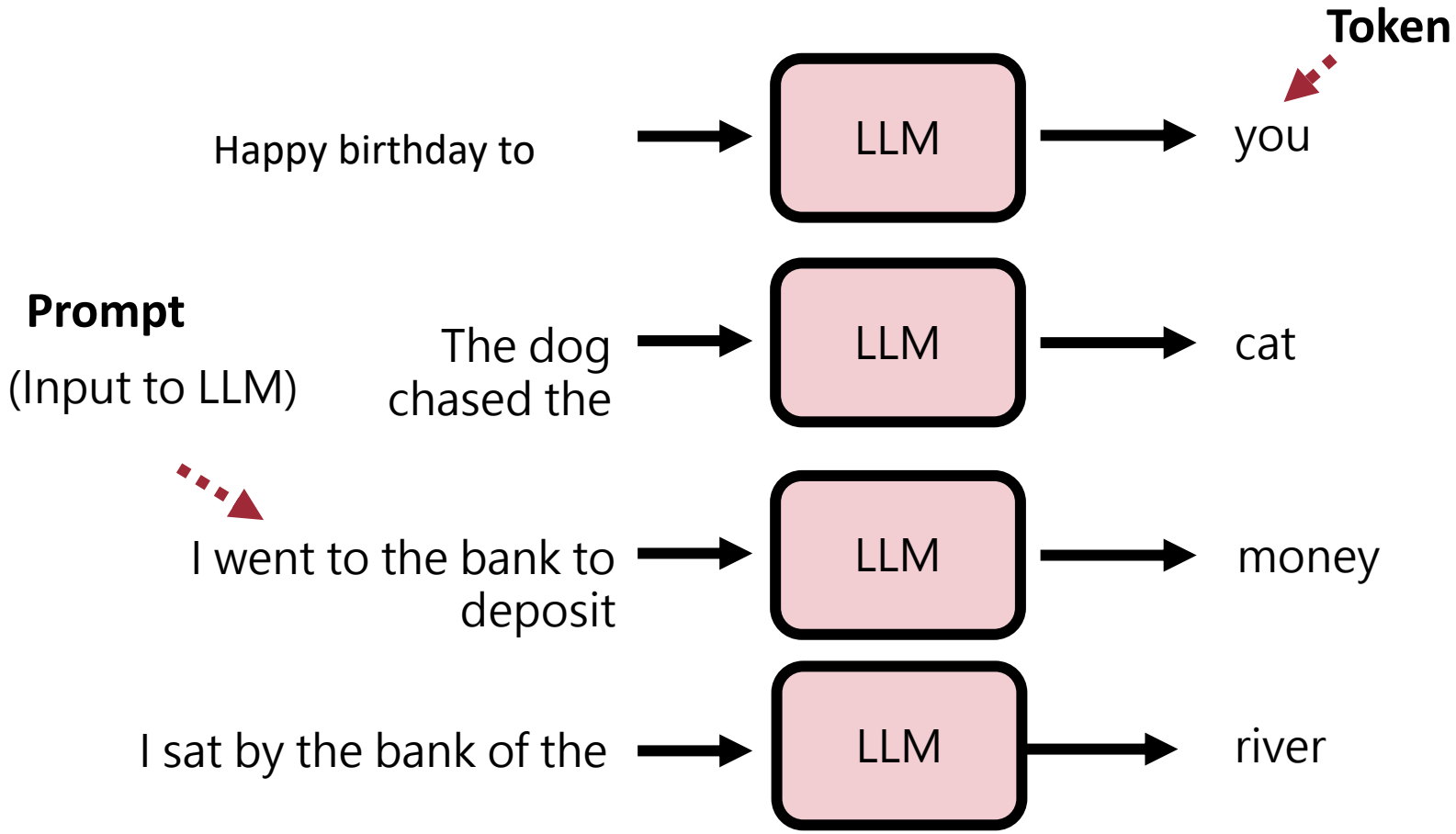
Generative Artificial Intelligence (Generative AI) refers to a class of artificial intelligence technologies designed to autonomously generate new content by learning patterns from large-scale datasets. These systems are capable of producing diverse forms of digital output, including text, images, audio, video, and computer code.

Generative AI
Key Characteristics

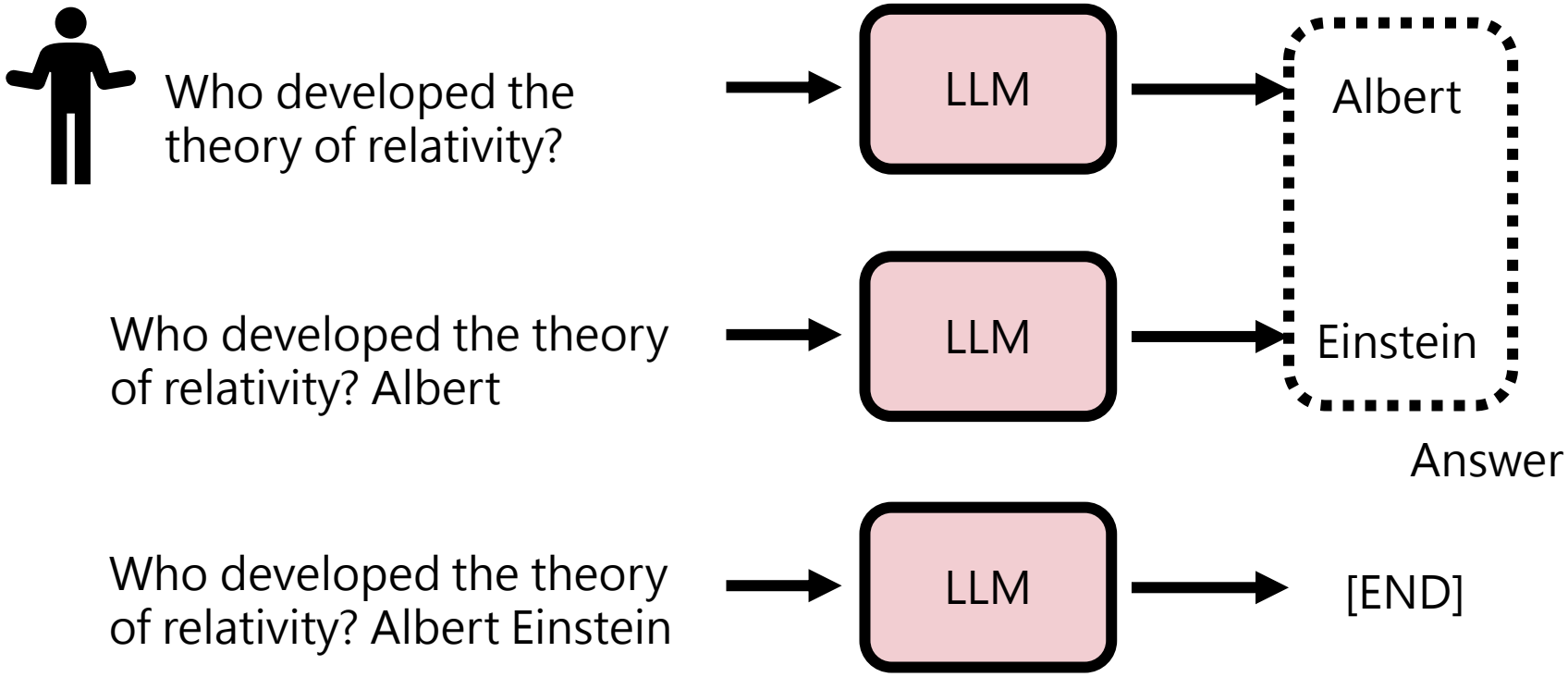
- **Content Generation Capability:**
  Generative AI systems not only interpret and analyze input data but also synthesize new content that reflects learned statistical and semantic patterns.
- **Computational Creativity:**
  By modeling complex relationships within data, these systems can emulate aspects of human creativity, producing outputs that are novel, coherent, and contextually relevant.
- **Data-Driven Learning:**
  Generative AI relies heavily on deep learning architectures trained on large datasets, such as Generative Adversarial Networks (GANs) and large language models (LLMs), enabling scalable and adaptive content generation.

ChatGPT, Gemini, Claude … Language Model Examples

Next-token prediction



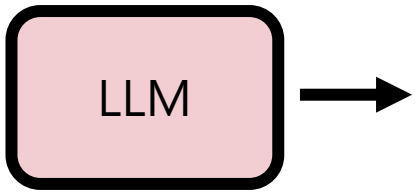
How a LLM answers ?



How a LLM Predicts

How LLM knows which token to choose

The airplane is flying in the …



Token probabilities

Token	Score
sky	0.72
air	0.15
clouds	0.10
자	0.0000…
ち	0.0000…
@	0.0000…
……	

Rolling the dice

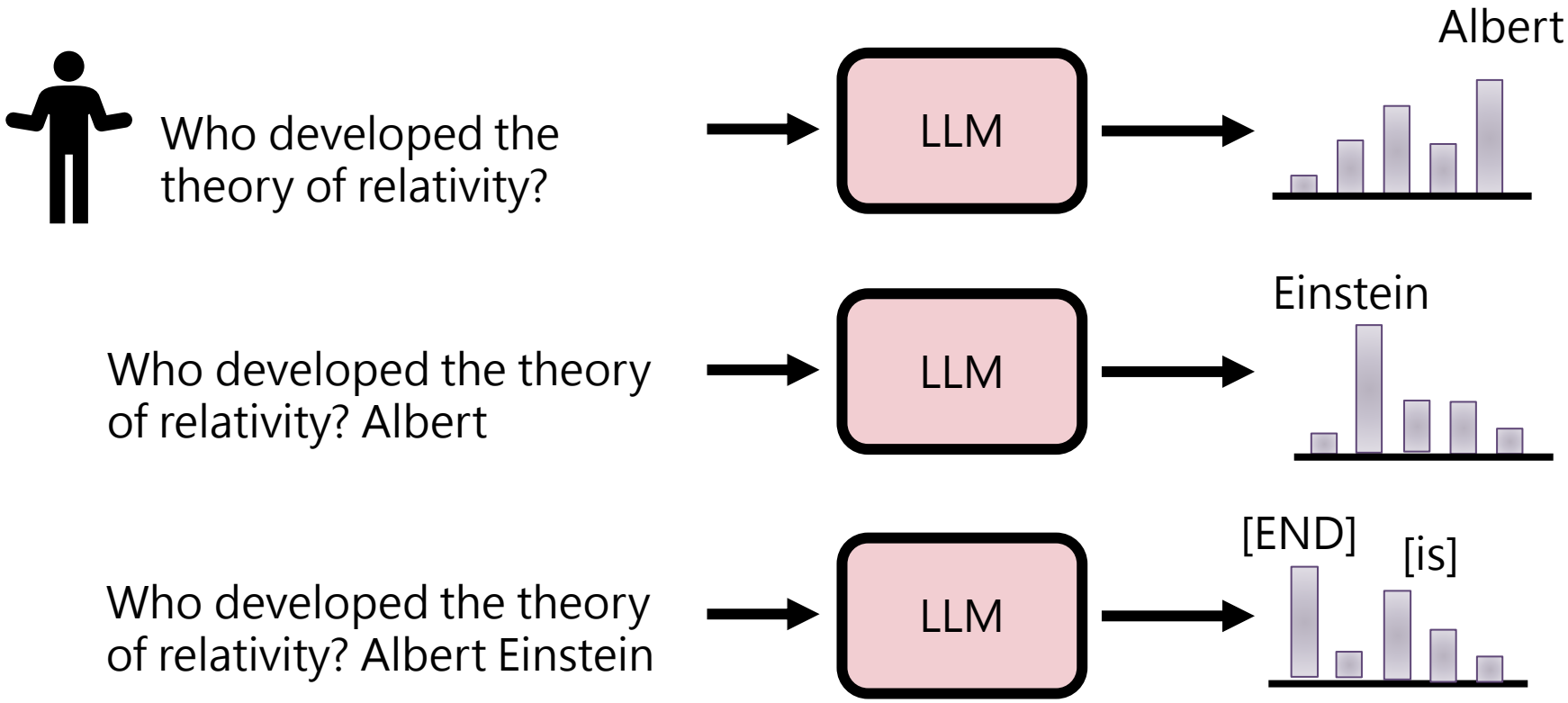
sky

72% probability

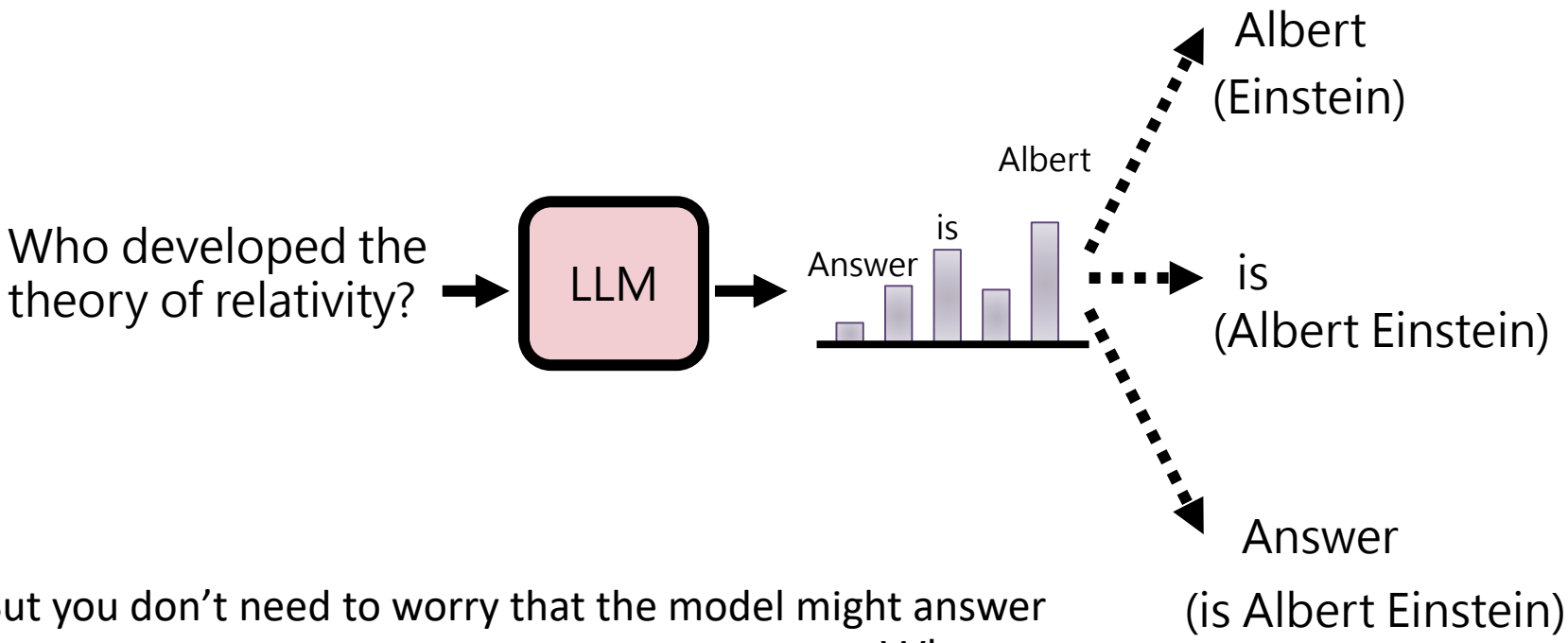
Vocabulary

Token needs to cover all output

How a LLM answers ? Revisit



You may get different answers given the same prompt



But you don't need to worry that the model might answer 'chicken sandwich,' because the probability that 'Who developed the theory of relativity?
' is followed by 'chicken' is extremely low.

Predicting Next Token is not an Easy Task

Language

Relatively Easy to Learn

Yellow___ → LLM → Noun (High probability)
Verb (Low probability)

World

Extremely Large

The boiling point of water is___°C → LLM → 100 (Hp)
0 (Lp)

At 0.5 atmospheres of pressure, the boiling point of water is __ °C → LLM → 100 should not be the answer with high probability

How LLMs are Trained
Public and Proprietary Data

Public Web Data

This is the largest and most diverse source.

•Websites (news, blogs, forums)

•Open knowledge bases (e.g., Wikipedia)

•Public discussions (e.g., Reddit)



Why it matters:

Provides broad coverage of language, topics, and real-world usage.

Licensed / Proprietary Data

Data that organizations obtain legally through agreements.

•Academic databases (e.g., Elsevier, Springer)

•Books, journals, and curated corpora

•Industry datasets

Why it matters:

Higher quality, more reliable, and often domain-specific (e.g., scientific, medical, legal).

How LLMs are Trained
Human-Created / Annotated Data

Data generated or refined by humans specifically for training.
Human-written examples and prompts
Labelled datasets for supervised learning

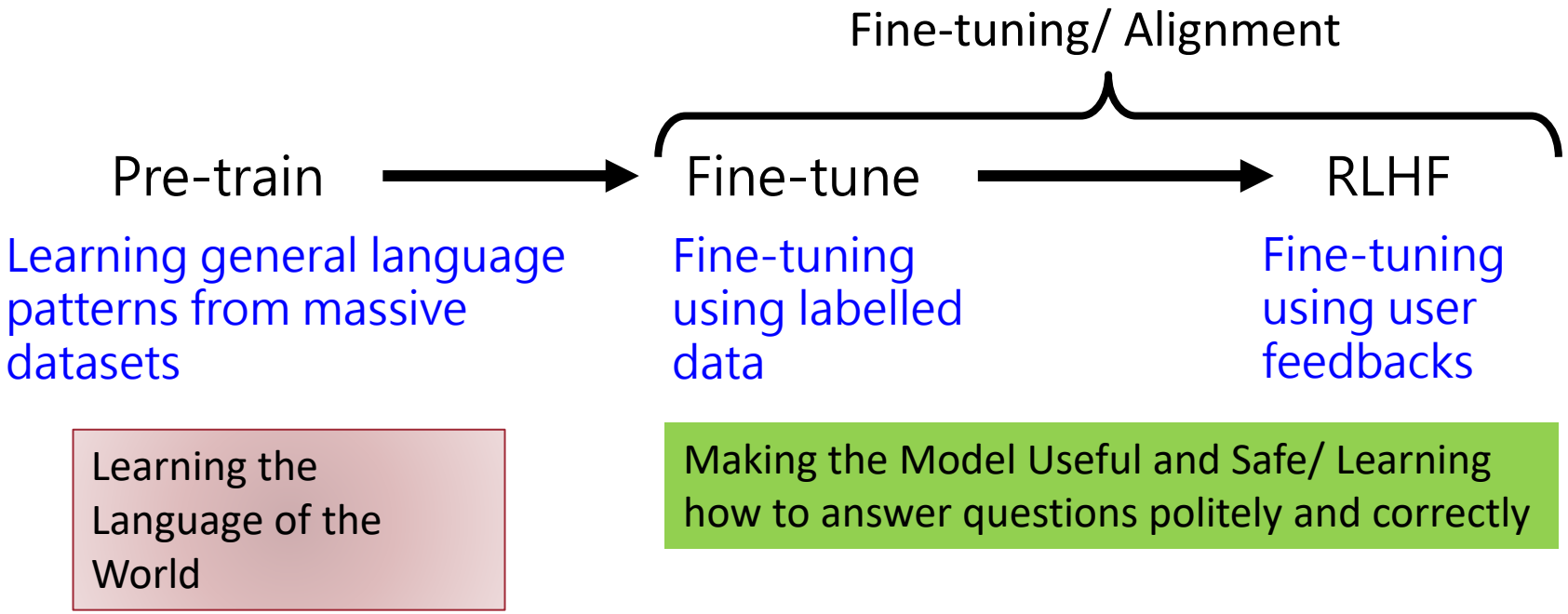


Feedback data (e.g., reinforcement learning with human feedback—RLHF)



Why it matters:
Improves accuracy, alignment, reasoning, and safety.

How LLMs are Trained?



Pretraining alone → knowledgeable but **unpredictable**
Fine-tuning alone → **too limited** without broad knowledge
Together → A model that is both **capable** *and* **aligned**

How LLMs are Trained
Pretraining

How it works

Objective: predict the next token (word/subword)

Training type: self-supervised learning

Data: large-scale mixture of:

- web text
- books
- articles
- code

What the model learns

- Grammar and syntax
- General knowledge
- Reasoning patterns (emergent)
- Statistical relationships between concepts

The model is **not taught facts explicitly**—it **absorbs patterns** from data.

How LLMs are Trained
Supervised Fine-Tuning (SFT)

How it works

- Humans write **input** → **ideal** output examples
- Model learns to imitate high-quality responses

Effect

- Improves clarity and instruction-following
- Makes responses more structured and useful

Example
Input: “Explain reinforcement learning”
Output: a clear, structured explanation

How LLMs are Trained
RL from Human Feedback(RLHF)

How it works

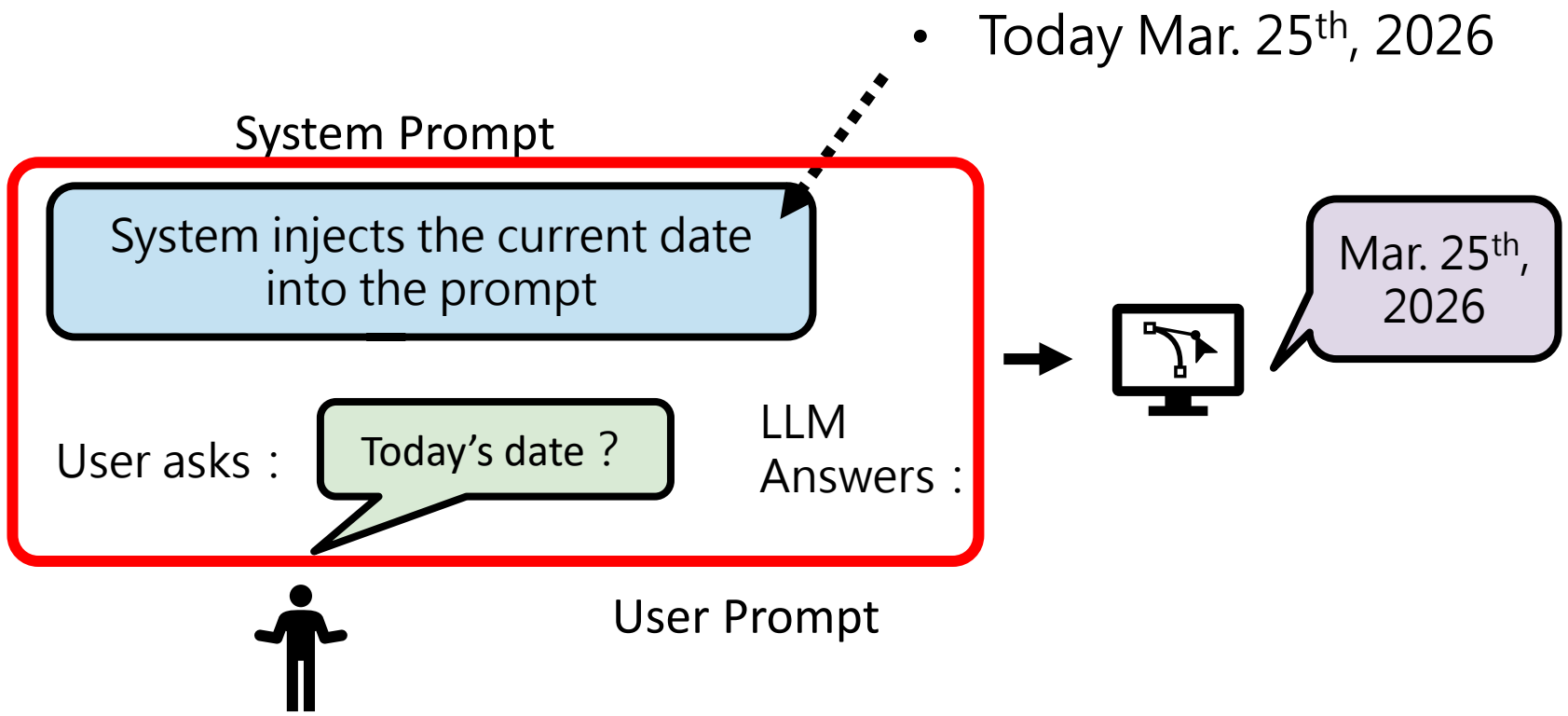
- Humans rank multiple model outputs
- A reward model is trained
- The LLM is optimized to produce preferred responses

Effect

- Reduces harmful or nonsensical outputs
- Aligns model with human values (helpfulness, honesty, safety)



Ensure Sufficient Inputs
Context Engineering



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Should I Trade QQQ Today?

“Should I trade QQQ today?”

Chatbot answer ≠ **Decision-making system**

Before acting, what evidence would you require?

- current data
- source quality
- calculation
- risk limit
- human approval

The User Builds the Conversation

User questions

What is the market environment?

What happened to QQQ recently?

What are the major news items?

Is it overbought or oversold?

What are today's risks?

Should I buy, hold, or sell ?

Chatbot synthesis

The model summarizes the provided information, compares arguments, explains uncertainty, and suggests possible actions.

Hidden issue:
the user decides every next question.

Why the Chatbot Is Useful

- **Summarize information**
- **Compare arguments**
- **Interpret qualitative context**
- **Explain uncertainty**
- **Generate alternatives**

LLMs add value when interpretation and synthesis matter.

But we still need to decide which facts, calculations, checks, and actions belong outside the model.

Use Software for Exact Work; Use the LLM for Judgment

Software should calculate	LLM should help interpret
Current price	Market narrative
Moving averages	News implications
RSI and volatility	Alternative explanations
Drawdown	Contradictory evidence
Portfolio exposure	Risk interpretation
Position size	Rationale in plain language

Reliable systems assign work to the component best suited to it.

Fluency Does Not Validate a Recommendation

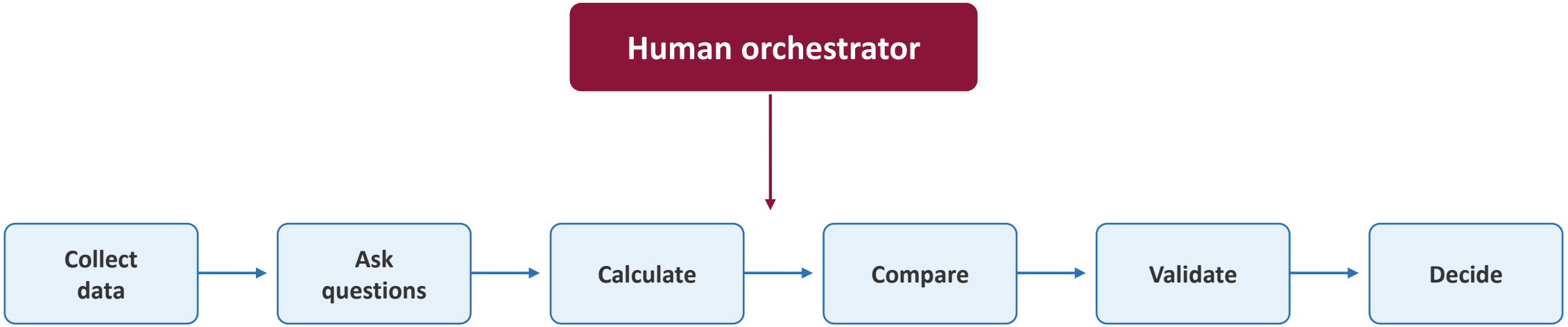
“QQQ looks oversold; this may be a buying opportunity.”

→

- ✓ What was observed?
- ✓ What was calculated?
- ✓ Which rule was applied?
- ✓ Which source and timestamp?
- ✓ Can it be reproduced?

A confident explanation can still be unsupported, inconsistent, or not reproducible.

The Human Is the Hidden Orchestrator



In the chatbot approach, the user decides what to inspect, which tool to use, and when to stop.

Class question: who is actually behaving like the agent here?

What Must a Dependable System Add?

Deterministic computation

Current observations

"Should I trade QQQ today?"

Persistent state

Validation

Constraints

Stopping conditions

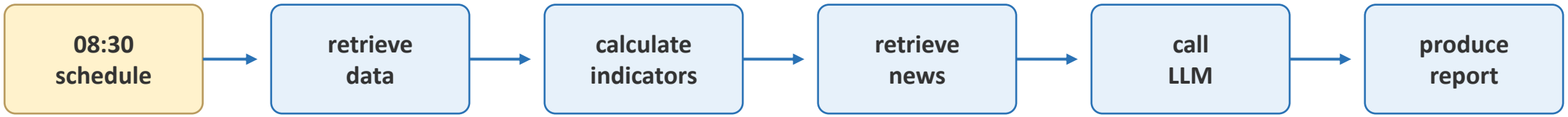


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A Workflow Executes Known Steps



The path exists before the run starts.

Predetermined orchestration is often enough for routine, repeatable work.

Fixed Automation Delivers Real Value

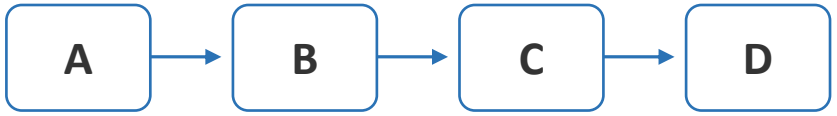


Faster | Repeatable | Inspectable | Easier to test | Less manual work

Agent autonomy is not required to improve speed, consistency, and repeatability.

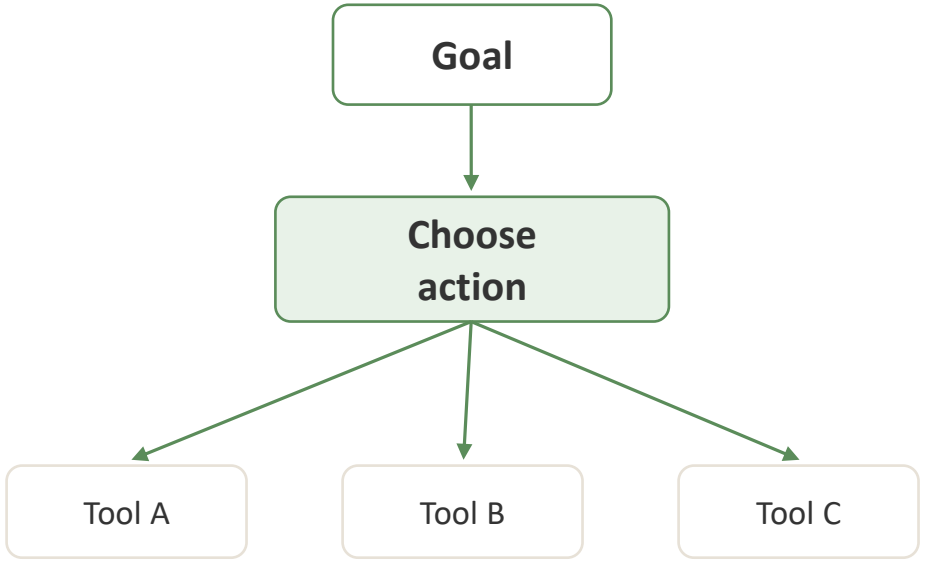
Workflow and Agent Differ in Who Chooses the Next Step

Workflow



Software chooses the sequence.

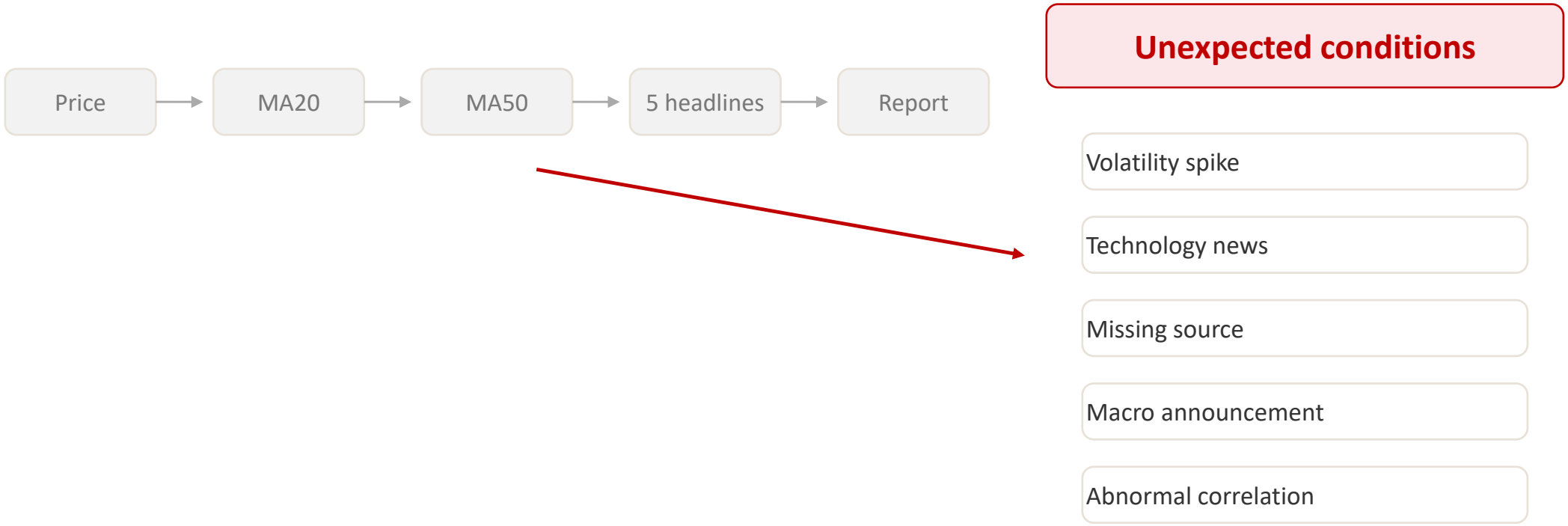
Agent



Model-directed action selection within boundaries.

The difference is control.

An Unusual Day Breaks the Routine



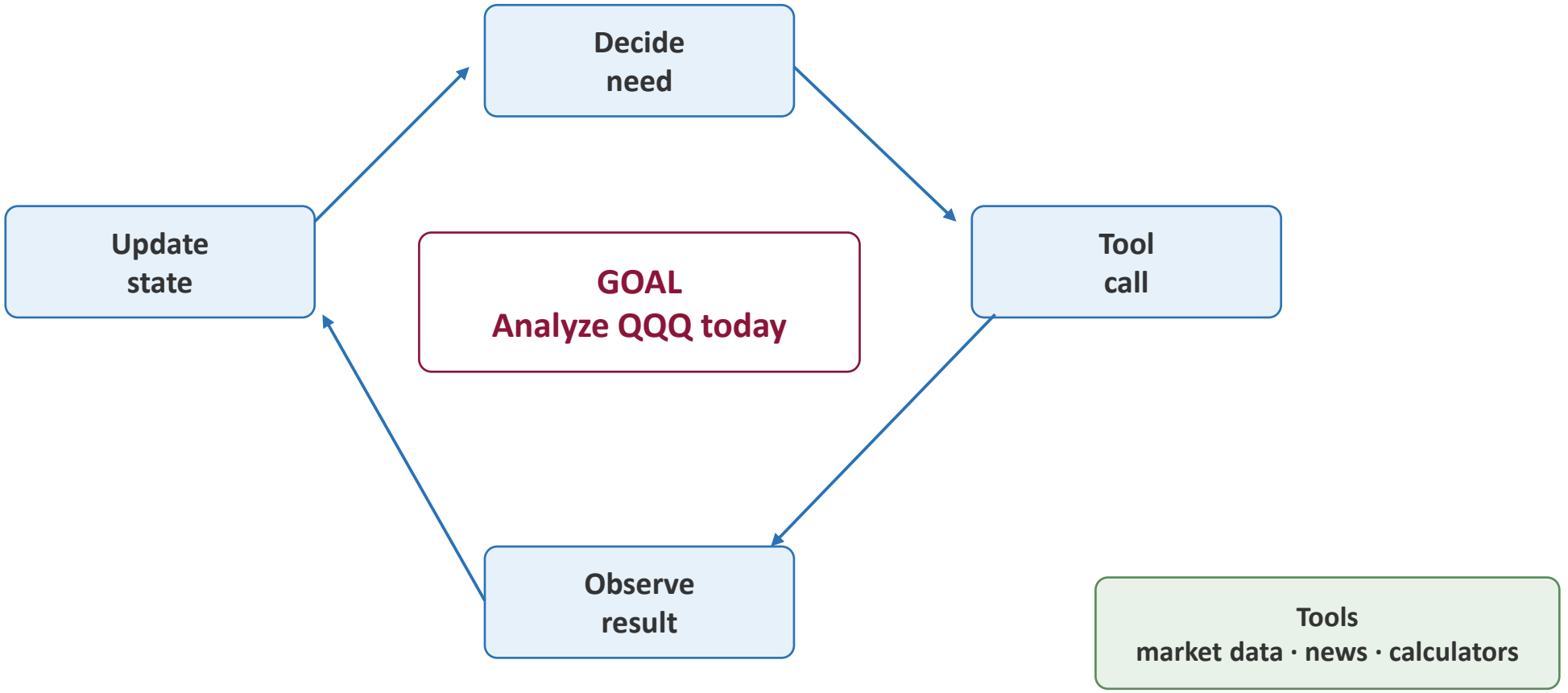
When assumptions fail, the system may need observations that were not planned in advance.

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A Single Agent Controls an Evidence-Gathering Loop



The agent selects tools, observes results, revises its plan, and stops under explicit conditions.

Five Components Give the Agent Its Structure

QQQ analysis agent

Control loop
what happens next

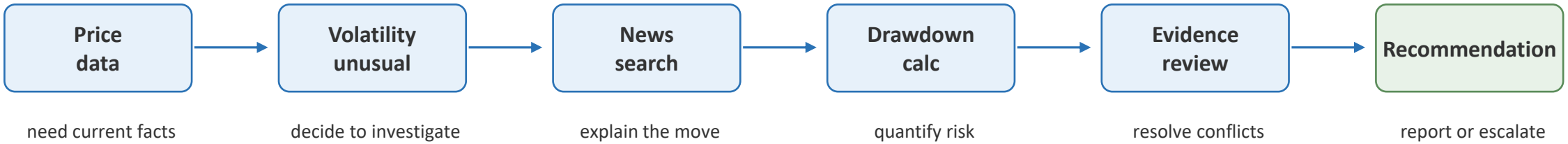
Model
reasoning
& judgment

Instructions
role, rules,
objectives

Tools
data, computation,
actions

State
progress & variables

One Run Builds Its Evidence Incrementally



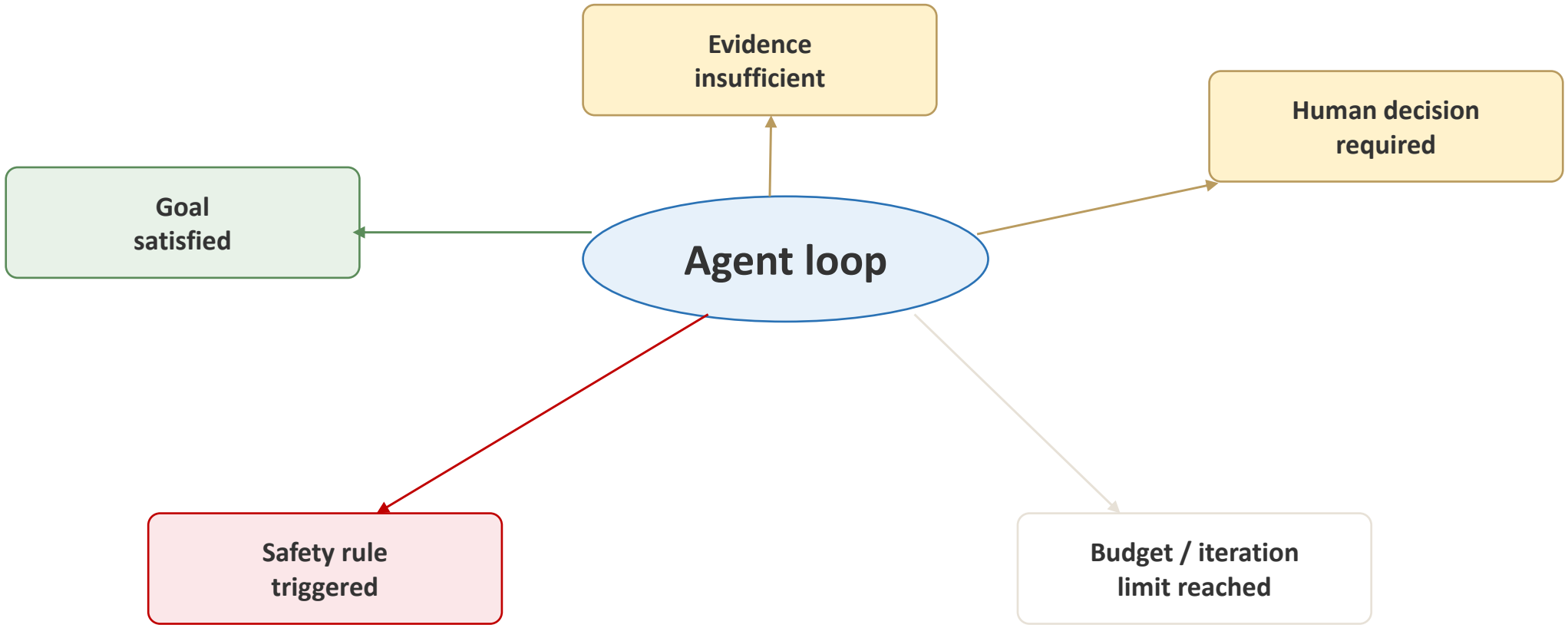
Later actions depend on observations gathered earlier in the run.

Context, State, and Memory Are Not Synonyms

Context	State	Memory
Current model call Prompt + observations passed now	Authoritative run progress Variables, tasks, and tool outputs	Retained across calls or runs Stable preferences or history
Example: latest tool result is included in the prompt.	Example: RSI=62; news searched; risk check pending.	Example: user requires human approval before trades.

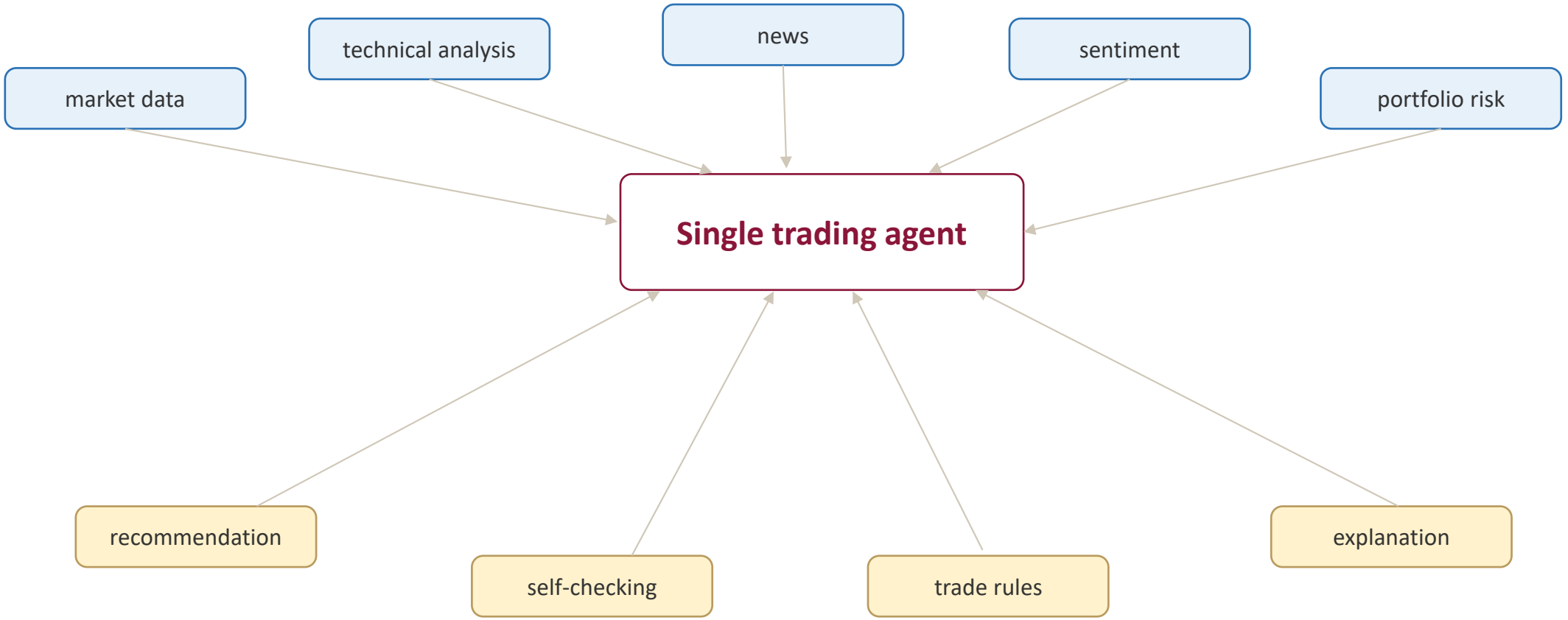
Reliable systems do not let prose history become the database.

Stopping Is Part of the Architecture



Completion, escalation, failure, and approval must be explicit outcomes.

One Agent Can Accumulate Too Many Responsibilities



A single agent becomes harder to instruct and evaluate when domains, tools, and success criteria collide.

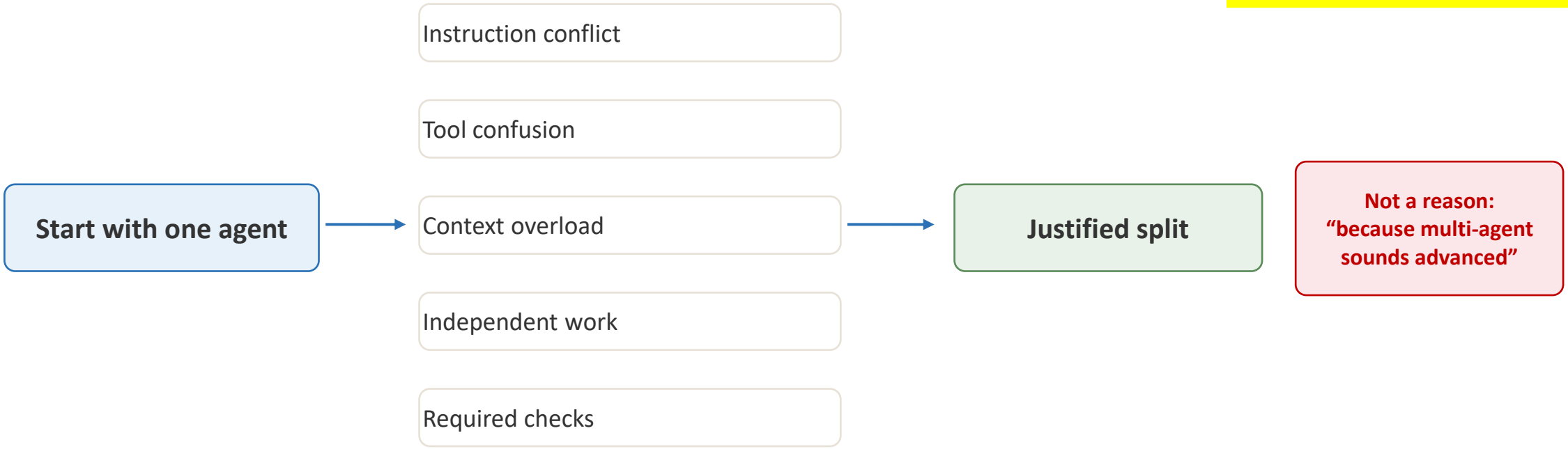
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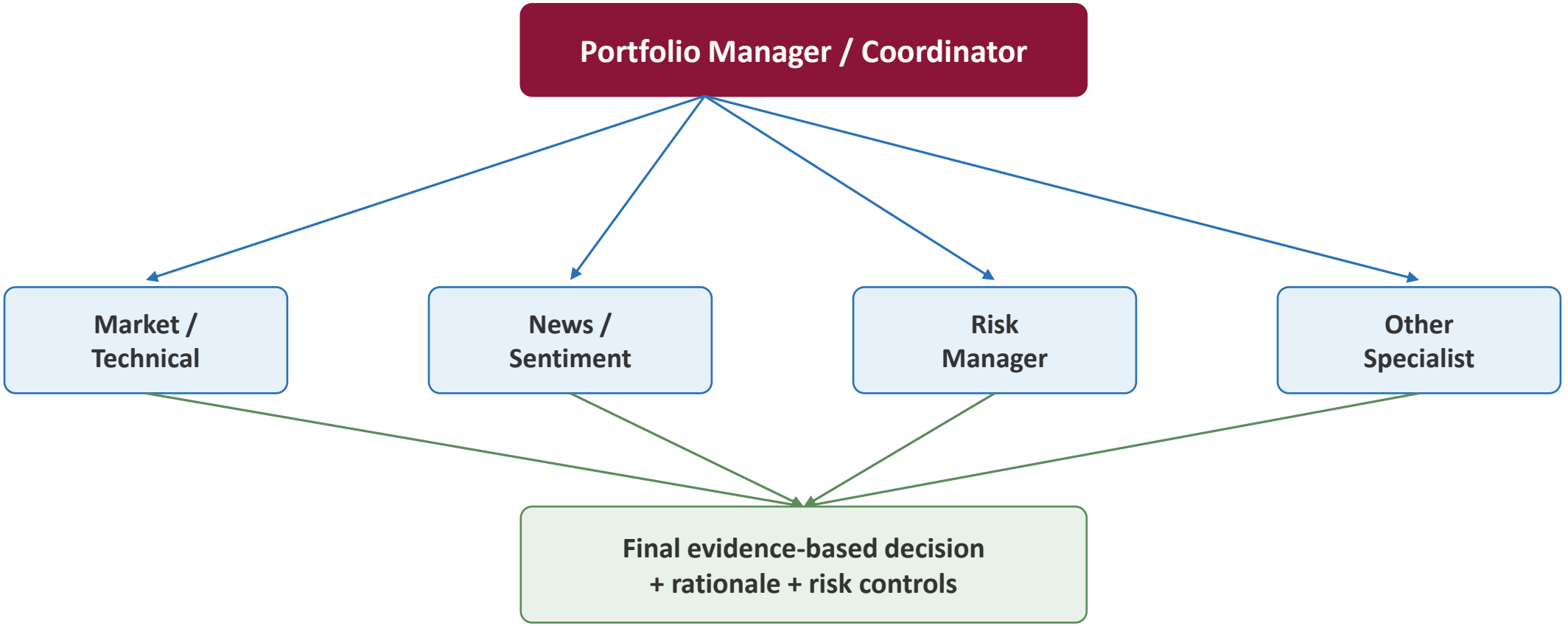
Split the System for a Specific Reason

Distribution of Control



Agent boundaries should solve a real engineering pressure.

A Manager Agent Preserves One Point of Control



The coordinator delegates bounded analyses and synthesizes their evidence.

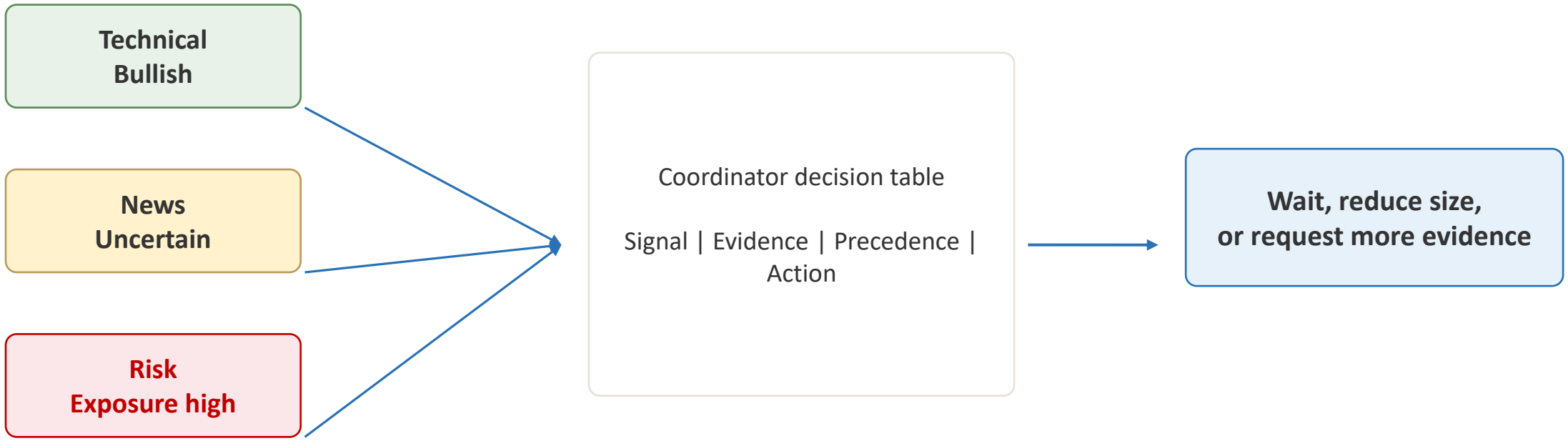
Each Specialist Owns a Different Evidence Domain

Market / Technical	News / Sentiment	Risk	Manager
data, indicators, trend, volatility	events, macro context, sentiment	exposure, constraints, uncertainty	evidence plan and synthesis
Output: quantified market view	Output: evidence summary	Output: risk limits	Output: recommendation

Clear ownership reduces prompt conflict and makes component evaluation possible.

Keep trade execution outside the analysts' authority.

Conflicting Agents Are Expected



The coordinator should preserve and resolve disagreement, not average it away.

The More Agents The Better?

Potential benefits	New costs
Specialization	More API calls
Role-specific tools	Higher latency
Parallel work	Coordination logic
Independent review	Contradictory outputs
Component evaluation	Harder debugging

Use the simplest architecture that meets the requirement.



Outline

- Agenthood, Agency and Agentic AI
- Agent-Based Modeling (ABM)
- Methods for Agent-Based Modeling
- Generative AI and Large Language Models (LLMs)
- Case Study: Agentic Trading Systems
  - Task: Should I Trade QQQ Today?
  - Workflow: Automating Routine Tasks
  - Agent: Autonomous Decision-Making
  - Multi-Agent Systems: Specialization and Coordination

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